

Project 3

Kinematically-Redundant Planar Parallel Manipulator - RRR

Format: Group of two students; simulation-based project.

Motivation and context

Kinematic redundancy allows a robot to avoid singularities that would otherwise limit its workspace and degrade its accuracy. Indeed, the extra degrees of freedom can be used to select favorable configurations, avoid joint limits and obstacles, and maintain a well-conditioned Jacobian even near configurations that would be singular for a non-redundant robot.

A kinematically-redundant planar parallel robot is shown in Fig. 1. The robot is also shown in the online video https://youtu.be/_vp1ELEtDN4. Two 2-PRR legs are attached to an articulated platform by revolute joints. Each leg is able to control the position of its corresponding revolute joint. The articulated end-effector platform has a passive revolute joint. The robot is therefore kinematically redundant: the end-effector has an additional DOF related to the revolute joint.

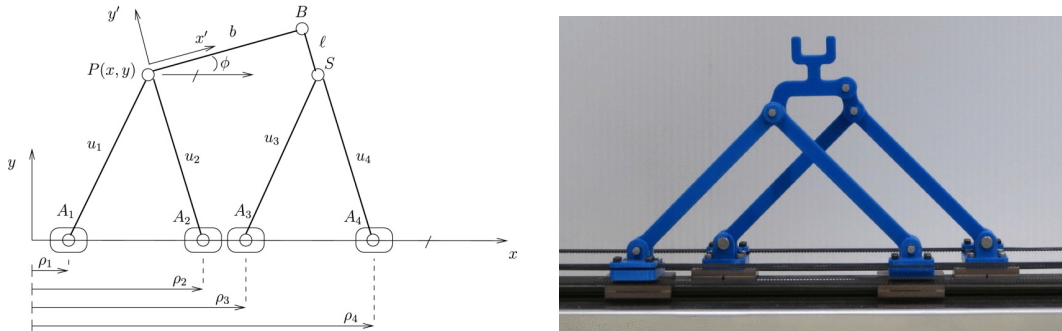


Figure 1: Kinematics architecture and prototype of the kinematically-redundant planar robot [1].

In this project, a similar case is considered. The 2-PRR leg mechanisms from [1] are replaced by 5-bar mechanisms with revolute joints, as depicted in Fig. 2.

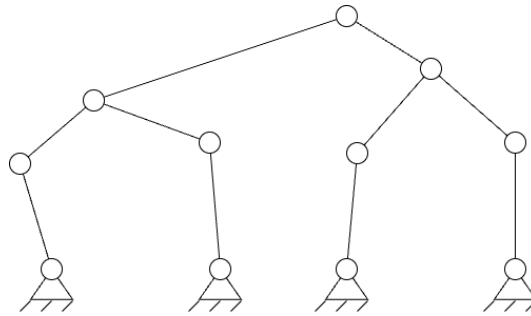


Figure 2: Considered architecture.

Tasks

Overall objectives. Perform a singularity analysis of the entire robot. Show that one of the platform links can undergo unlimited rotation with appropriate redundancy resolution strategies.

- **Compute analytical forward and inverse kinematics.** Hint: for the forward kinematics, solve first for the 5-bar legs, then for the reconfigurable platform.
- **Create a Simscape model of the robot.** Use this model in order to verify the kinematics equations. Geometrical parameters must be chosen such that unlimited rotation is feasible.
- **Analyze the singularities.** Use results from Grassmann geometry to provide a geometrical interpretation. Hint: read paper [1].
- **Propose a singularity avoidance strategy.** This strategy should allow the end-effector to undergo unlimited rotation.
- **Design a trajectory in augmented task space.** This trajectory should illustrate the unlimited rotation capability.
- **Determine joint torques.** Consider quasi-static assumption with mass associated to the end-effector and gravity. Are the simulation results consistent with the singularity analysis?

Available documents

Paper [1] can be accessed online.

Expected outcome

Students should obtain the forward and inverse kinematics and the Jacobian matrices of the kinematically-redundant robot. They should develop a singularity avoidance strategy that enables unlimited rotation.

Submit a report, and a short video (screen recording) with Simscape illustrating unlimited rotation.

References

- [1] C. Gosselin, T. Laliberté, and A. Veillette, “Singularity-free kinematically redundant planar parallel mechanisms with unlimited rotational capability,” in *IEEE Transactions on Robotics*, 2015, doi: 10.1109/TRO.2015.2409433.